



Experimentally testing Bragg's reflection using
Bremsstrahlung, and measuring the absorption of
X-Rays using Aluminium plates

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August 2024

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1 Introduction

X-rays is unique within the electromagnetic spectrum, due to its vast application within human activities. Most notably, it is used within the medical field for imaging within the human body. However, today's experiment revolves around 2 phenomena closely attached to electromagnetic radiation. A materials' ability to absorb electromagnetic radiation, and the Bragg's reflection phenomena.

The reason x-rays are used within this experiment is due to another phenomena known as "Bremsstrahlung". This shall be explained with more depth within the "Background" section. Bremsstrahlung provides an easy method for generating x-rays specifically, but decelerating electrons.

2 Background

Bremsstrahlung translates from German into "braking radiation"[1]. When electrons are travelling at high speeds, and are then suddenly decelerated by strong electric fields from atomic nuclei of a certain substance, the energy is converted from kinetic energy into high frequency photons. Specifically, x-rays are generated from this interaction occurring. Phenomena such as cosmic rays have been attributed to Bremsstrahlung.

Bragg's law has to do with the lattice structure of some crystal structures, such as NaCl (the substance used within this experiment). When electromagnetic radiation is incident on a substance, the angle of reflection will be the same as the angle of incidence. However, due to the nature of crystal structures, there are "branches" of the crystal structure that the electromagnetic radiation may be incident on, that differ by a length of d by the branch atop[2].

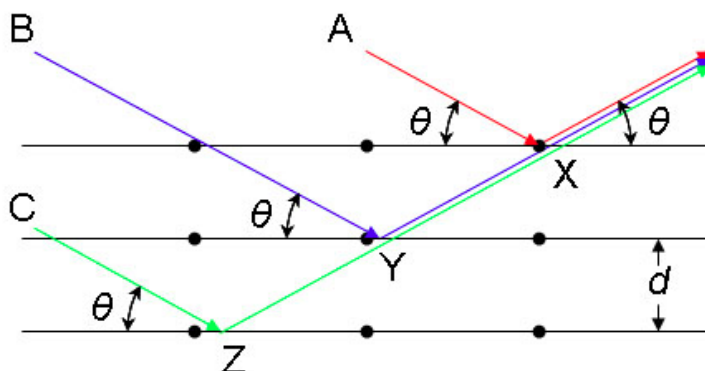


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When the path difference is equal to some whole number, then when the two individual electromagnetic waves interact with one another again, they will interfere construc-

tively [2].

Regarding absorption of x-rays by various materials (also known as attenuation of x-rays), like mentioned before, it has vast application within the medical field. It is this phenomenon where materials absorb x-rays that create the images one sees with x-ray machines in hospitals and such.

There are three ways in which x-rays are absorbed by materials. Compton scattering, photoelectric effect and pair-production absorption [3].

Compton scattering

Compton scattering refers to the process whereby a photon is absorbed by an electron, and is then subsequently emitted from the electron and another, lower energy photon is emitted as well. This process occurs when an electromagnetic wave collides with an electron, and the energy of the electromagnetic wave exceeds the binding energy of the electron, therefore ionising it. The remaining energy is then transformed back into a photon with less energy [3].

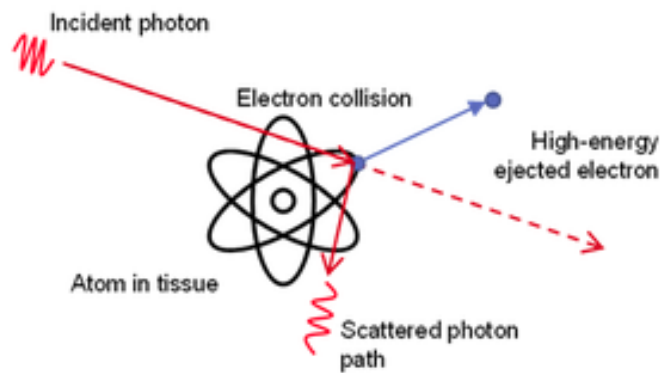


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Photoelectric effect

The photoelectric effect, similar to the Compton scattering effect, ejects electrons out of their atomic shells and ionizes the electrons. When an electromagnetic wave is incident upon an electron, and the electron fully absorbs the photon and the photon exceeds the binding energy of the electron, then the electron is emitted. The difference between this and Compton effect is that no residual energy is transformed into a photon with lesser energy. The electron fully absorbs the photon [3].

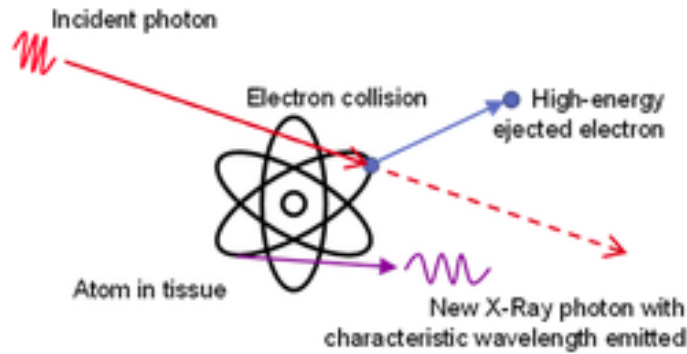


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Pair-production absorption

Pair-production absorption involves the photon interacting with the nucleus, leading to the photon transforming into an electron-positron pair [3]. A positron is an electron, but instead of having a negative charge, it has a positive charge. Once the transformation from photon to pair of particles has occurred, the particle pairs will travel a certain distance before colliding and annihilating one another. This then leads to the creation of 2 photons that fly off in opposite directions.

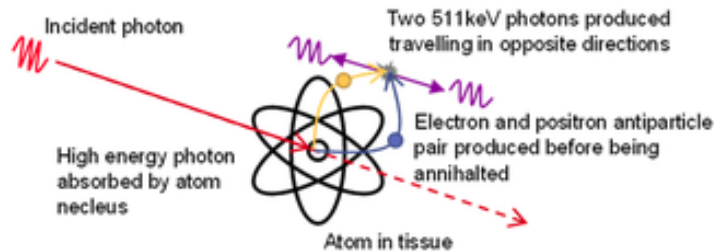


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3 Mathematical background

Regarding Bragg's law, when an x-ray is incident on a crystalline structure, it is then reflected at an angle θ . However, if a wave of x-ray is incident on the bottom branch (relative to the initial one), and the distance is an integer value ($d = 1, 2, \dots$), then constructive interference will occur. These phenomena can be described mathematically using the following equation:

$$2d\sin(\theta) = n\lambda \quad (1)$$

Where d is the distance between the Bragg plates (or branches as referred to earlier), λ refers to the wavelength. In this experiment, the wavelength will be calculated. In order to achieve this, the distance d must be derived. This can be derived theoretically using the density of the crystal and assume the crystal has a lattice structure. The mass of an individual molecule of NaCl is:

$$\frac{N}{M}$$

Where N is Avogadro's number and M is the molar mass of NaCl. As the atom is diatomic, the volume of the NaCl can then be described as:

$$\frac{2pN}{M}$$

Thus the distance between the Bragg plates can then be described with the value 0.282nm.

Regarding the X-ray absorption experiment, the intensity of an electromagnetic wave that is incident upon a medium that travelled a distance x is described by:

$$I = I_0 e^{-\mu x} \quad (2)$$

Where μ is the linear absorption coefficient, unique to each material. In this experiment, we shall be testing the validity of this equation by calculating the μ value.

4 Procedure

This procedure shall be divided into two parts. One focusing on the absorption of x-rays by various materials, and the other focusing on the testing of Bragg's law, utilising a NaCl (table salt) crystal structure.

4.1 X-ray absorption

To start, first ensure that the device used to create Bremsstrahlung can safely contain the x-rays being emitted. X-rays, as stated in the background, can ionize electrons. This includes the particles that constitute one's being. Proceed cautiously.

For this section, aluminium plates of varying thickness are needed, and a mechanism capable of holding them. Ensure that a detector is placed behind the plates in order to detect x-rays that pass through.

1. Before starting the machine, ensure all safety measures are met.
2. Slot one of the aluminium disks into the holder.
3. Close the Bremsstrahlung device. Ensure that it is safely closed.
4. Read the counting rate the device measures and record the results.
5. Repeat this process for all other aluminium plates that will be measured. Record the data similarly

4.2 Bragg's law experiment

This experiment will make use of the same device used in the x-ray absorption experiment. Only that the x-rays being deflected by the sodium crystal will be measured instead of how much x-rays pass through a substance.

The x-ray measuring device should be placed on a device such that it can rotate around a certain circumference as to measure the angles at which the x-ray waves are being deflected.

1. Place the sodium crystal in the centre of the device, such that the x-rays are fired directly towards the crystal.
2. Place the x-ray deflection measuring device such that it is in line with the sodium crystal and the source of x-rays.
3. Once everything is in place, close the dome over the system. Once again, ensure that it is closed safely.
4. Begin the Bremsstrahlung process, and begin slowly rotating the x-ray detection device. Ensure to move the device at a constant rotational velocity. Within this experiment, the detection device was on a motor that ensured a constant velocity

5 Results

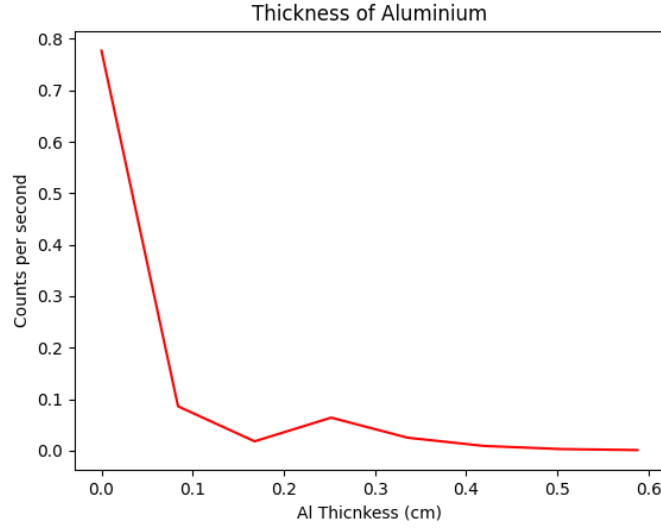
The results section shall also be divided into two sections. The first focusing on the X-ray absorption experiment, and the second focusing on the Bragg's law experiment.

5.1 X-ray absorption results

Note that the thickness at 0 shall be used as the I_o will be used

Aluminium Thickness (cm)	Counts per second
0	0.777
0.084	0.086
0.168	0.018
0.252	0.064
0.336	0.025
0.420	0.009
0.504	0.003
0.588	0.001

Table 1: Aluminium thickness of plate compared against counting rate in seconds



Graph 1: Aluminium thickness of plates against counts of x-rays per second

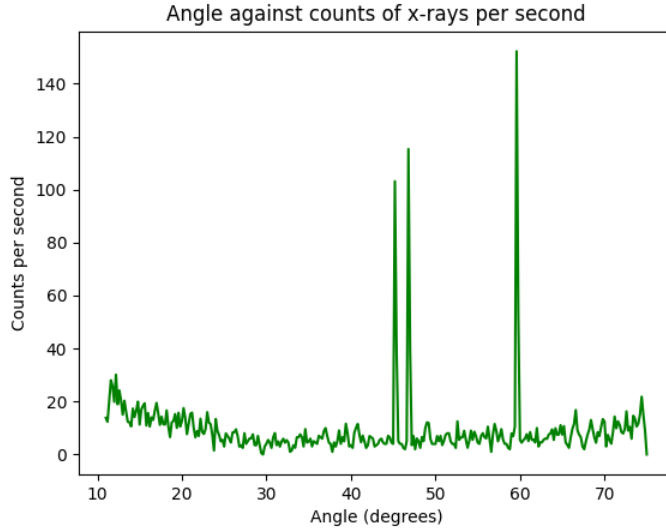
For the mathematical calculations, we must first make μ the subject of the equation. The equation then becomes:

$$\mu = -\frac{1}{x} \ln\left(\frac{I}{I_o}\right) \quad (3)$$

The calculations then yield the following values for μ :

$$\mu_1 = 26.203 \mu_2 = 22.411 \mu_3 = 9.910 \mu_4 = 10.228 \mu_5 = 10.615 \mu_6 = 11.025 \mu_7 = 11.319$$

5.2 Bragg's law experiment results



Graph 2: Graph of angle of detection device against x-ray's being emitted from NaCl crystal

For the mathematical calculations, we note that the graph peaks at around $\pm 45^\circ$, $\pm 46^\circ$ and $\pm 59^\circ$. Using these angles we find:

$$\lambda_1 = 0.399nm \lambda_2 = 0.405nm \lambda_3 = 0.483nm$$

6 Discussion

This discussion, similar to the results, will be divided into two parts. The first part discusses the results of the x-ray absorption experiment, and the second part discusses the results of the x-ray deflection occurred through the sodium crystal.

Regarding the x-ray absorption, the data does show a clear inverse relationship between thickness of a material and the counts of x-ray that penetrate the material. However, there are irregularities in the data that must be addressed. In the first section of the data, it follows a normal and expected trend. However, when the thickness increases past 0.2cm, there appears to be an increase in amount of x-rays that penetrated through. This could be attributed to errors with either the detection by the detector or the individuals handling the equipment. The overall data trend remains the same. When examining the μ values, while at first does change drastically, the value then settles upon a value of ± 10 . The change in values could again be attributed to error in mathematical use or in experimental procedure.

Regarding the x-ray diffraction experiment, although there are numerous peaks within the graph, there are 3 very prominent peaks, predominantly at $\pm 45^\circ$, $\pm 46^\circ$ and $\pm 59^\circ$.

These peaks do not stretch over larger angles, however. The x-rays peak there, and then quickly dissipate into smaller counts again. With the calculated wavelengths, the values seem to fit well into the accepted range of x-ray wavelength values. Though it must be noted that the gap between the last peak and the initial two is quite large. This again could be attributed to inconsistent use of machinery.

7 Conclusion

The aim of this experiment was to determine the penetrative characteristics of x-rays using aluminium plates, and to also measure Bragg's law using a sodium chloride crystal. Regarding the absorption experiment, a clear relationship was shown with how counts of x-rays decreased when the thickness of the plates increased (even with some data inconsistencies therein). These results can then be used further in application of x-ray technology (as in the medical field). If further tests are run using different materials of different plates, perhaps several materials of several thicknesses can then be applied and optimised depending on the field of application. When examining the μ values, although there were inconsistencies within the calculations, it still could point to the equation at use to be applicable. This could then be used further in other fields to improve optimization.

Regarding the x-ray diffraction (Bragg's law) experiment, this experiment also proved successful in finding the angles at which the x-rays constructively peaked throughout the trial. If the same method was used for a different crystalline structure, then perhaps different angles could be discovered. This, again, has vast applicable knowledge. Especially when diffracted x-rays are needed, or x-rays of a specific intensity are needed. Regarding the x-ray wavelengths calculated, it calculated them fairly accurately as all x-ray values fell into the accepted range of x-ray wavelengths. This experiment could then be used to further study lattice structures of other molecular structures that are similar to Sodium Chloride.

8 References

References

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- [3] StudySmarter. *Absorption of X-Rays*. StudySmarter. Accessed: 2024-08-09. 2024. URL: <https://www.studysmarter.co.uk/explanations/physics/medical-physics/absorption-of-x-rays/>.